

# Learning Non-Uniform Step Sizes for Neural Network Quantization

Shinya Gongyo<sup>1</sup>, Jinrong Liang<sup>2</sup>, Mitsuru Ambai<sup>1</sup>,  
Rei Kawakami<sup>2</sup>, and Ikuro Sato<sup>1,2</sup>

<sup>1</sup> [Denso IT Laboratory](#), Tokyo, Japan

<sup>2</sup> [Institute of Science Tokyo](#), Japan

Corresponding author: [gongyo.shinya@core.d-itlab.co.jp](mailto:gongyo.shinya@core.d-itlab.co.jp)

**Abstract.** Quantization of neural networks enables faster inference, reduced memory usage, and lower energy consumption, all of which are crucial for deploying AI algorithms on devices. However, quantization may degrade performance compared to full-precision models as precision decreases. While prior research has primarily focused on uniformly quantizing network weights and activations, capturing the long-tail distributions of these quantities imposes a challenge. To address this issue, this paper introduces a non-uniform learned step-size quantization (nuLSQ) approach. It optimizes individual step sizes for quantizing weights and activations. Evaluations on CIFAR-10/100 and ImageNet datasets, using ResNet, MobileNetV2, Swin-T, and ConvNeXT with 2-, 3-, and 4-bit precisions, demonstrate that nuLSQ outperforms other quantization methods. The code is available at <https://github.com/DensoITLab/nuLSQ>.

**Keywords:** Quantization · Compression · Real-time inference

## 1 Introduction

Deep learning has improved model performance across a range of applications such as image classification [8, 14, 51], object detection [47, 48], speech synthesis [41], and language translation [52]. In these applications, large over-parameterized models can generally achieve high performance [60], but they require huge computational costs during inference. Compact models that solve the same tasks with higher inference speed, lower memory footprint and lower energy consumption are in high demand especially for low-end edge devices.

Aiming at speeding up inference with low memory consumption and little performance loss, methods of neural network compression and acceleration have been introduced, such as knowledge distillation [3, 16, 24, 29, 56, 59], network pruning [15, 19, 35], and network quantization [5, 6, 9, 10, 30, 32, 45]. This work focuses on neural network quantization given its capability of reducing the number of floating-point operations and memory footprint by replacing real-valued weights and activations with integer-valued ones. Although a quantized network has an inherent problem of quantization error, training of quantized networks with little performance loss is becoming feasible [6, 9, 10, 22, 61].